Project - Low Level Design

on

(Sports Planning Assistant Agent)

Course Name: Agentic - AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Introduction

1.1 Scope of the Document

This document presents the design and architecture of the Sports Planning Assistant Agent, an intelligent system developed to assist users in planning and organizing sports-related activities efficiently.

The system leverages agent-based architecture and Large Language Model (LLM) APIs to automate complex planning tasks such as scheduling, task decomposition, and decision-making.

The scope includes:

- System architecture
- Application workflow
- Component design
- API usage
- Key design considerations

1.2 Intended Audience

This document is intended for:

- Software developers and engineers
- AI/ML practitioners
- Academic evaluators and project reviewers
- Students studying intelligent systems

1.3 System Overview

The Sports Planning Assistant Agent is an AI-driven system that helps users organize sports events and activities by breaking down high-level goals into structured plans.

The system processes user queries such as:

- "Plan a cricket match this Sunday"
- "Generate match statistics workflow"

It then:

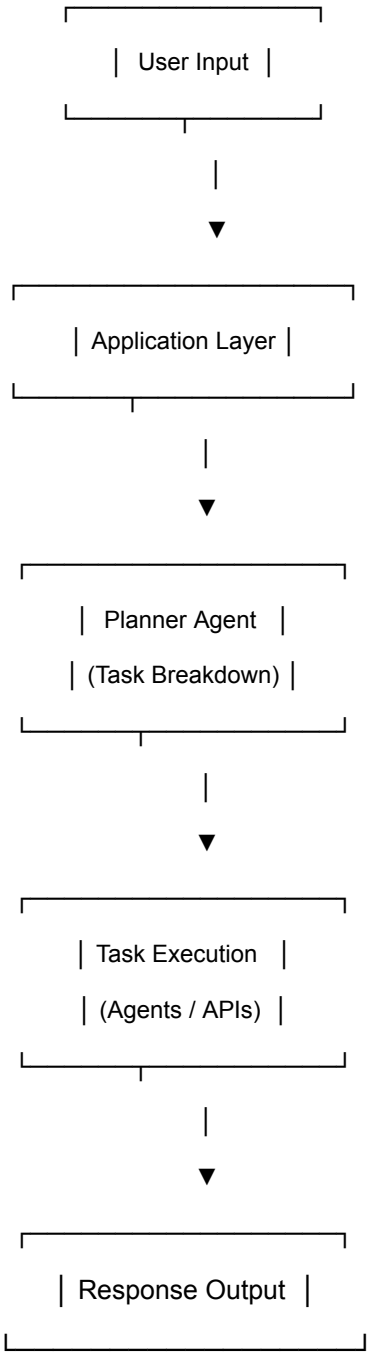
- Decomposes the task into smaller steps
- Determines execution order
- Utilizes available tools
- Generates a detailed plan or response

The system follows a modular agent-based design, where different agents handle specific responsibilities such as planning, execution, and information retrieval.

2. System Design

2.1 Application Design

The system is designed using a modular architecture, ensuring flexibility, scalability, and ease of maintenance.



Architecture Type:

- Agent-Based Modular System

Technology Stack:

- Programming Language: Python
- AI Framework: CrewAI / LangChain
- LLM Integration: OpenAI or similar APIs
- Configuration Management: Environment variables

Design Approach:

- Separation of concerns through agents
- Reusability of components
- Extensibility for future enhancements

2.2 Process Flow

The system follows a structured workflow to process user requests:

Step-by-Step Flow:

1. **User Input:**
   The user provides a high-level goal (e.g., "Plan a football tournament").
2. **Input Processing:**
   The application receives and interprets the input.
3. **Planner Agent Activation:**
   The planner agent analyzes the goal and decomposes it into sub-tasks.
4. **Task Decomposition:**
   Example:

   - Select venue
   - Schedule matches
   - Organize teams
   - Generate statistics

5. **Resource Validation:**
   The system checks available tools and APIs required for execution.
6. **Task Scheduling:**
   Tasks are arranged in a logical sequence considering dependencies.
7. **Execution**:
   Each task is processed using appropriate agents or APIs.
8. **Response Generation:**
   The system compiles results into a structured output.
9. **Output Delivery:**
   The final response is presented to the user.

2.3 Information Flow

The system processes and transfers information through multiple stages:

User Query

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Application Layer

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Planner Agent

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Task Decomposition

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Execution Modules / APIs

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Processed Data

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Final Output

Data Characteristics:

- Input: Natural language text
- Processing: Structured intermediate tasks
- Output: Organized plan or response

Key Aspects:

- No long-term storage required
- Real-time processing
- Stateless interaction

2.4 Components Design

The system consists of several key components:

1. User Interface

- Accepts user queries
- Displays results
- Can be CLI or simple web interface

2. Core Application Layer

- Acts as the central controller
- Manages flow between components
- Handles input/output operations

3. Planner Agent

- Core intelligence of the system
- Breaks high-level goals into sub-tasks
- Determines execution sequence
- Handles dependencies

4. Task Execution Agents

a) Scheduling Agent

- Creates schedules for sports events

b) Information Retrieval Agent

- Fetches required data from external sources

c) Recommendation Agent

- Suggests improvements and optimizations

5. Tool Integration Layer

- Interfaces with external APIs
- Ensures smooth data exchange

6. LLM Processing Unit

- Generates intelligent responses
- Assists in reasoning and planning

7. Configuration Module

- Manages API keys and environment variables
- Ensures secure handling of credentials

2.5 Key Design Considerations

1. Scalability

The system is designed to support additional agents and features without major restructuring.

2. Performance

Efficient task execution ensures minimal latency in generating responses.

3. Security

- API keys are securely stored
- Sensitive data is not exposed

4. Modularity

Each component operates independently, allowing easy updates and maintenance.

5. Extensibility

New functionalities such as booking systems or live tracking can be added in the future.

6. Reliability

The system ensures consistent performance even when handling complex queries.

7. Dependency Handling

Tasks are executed in logical order to maintain correctness

2.6 API Catalogue

API Name	Purpose	Usage
LLM API	Natural language understanding and response generation	Core processing
Search API	Retrieve real-time information	Data fetching
Sports Data API (optional)	Provide match-related data	Statistics generation
Weather API (optional)	Check weather conditions	Planning assistance

3. References

- Documentation of Large Language Models (LLMs)
- CrewAI / LangChain Framework Documentation
- Research papers on Agent-Based Systems
- Sports event management concepts

Conclusion

The Sports Planning Assistant Agent demonstrates the use of AI in automating complex planning tasks. By leveraging agent-based architecture and LLM capabilities, the system efficiently converts high-level user goals into structured execution plans.

The design ensures scalability, flexibility, and extensibility, making it suitable for real-world applications in sports management and beyond.